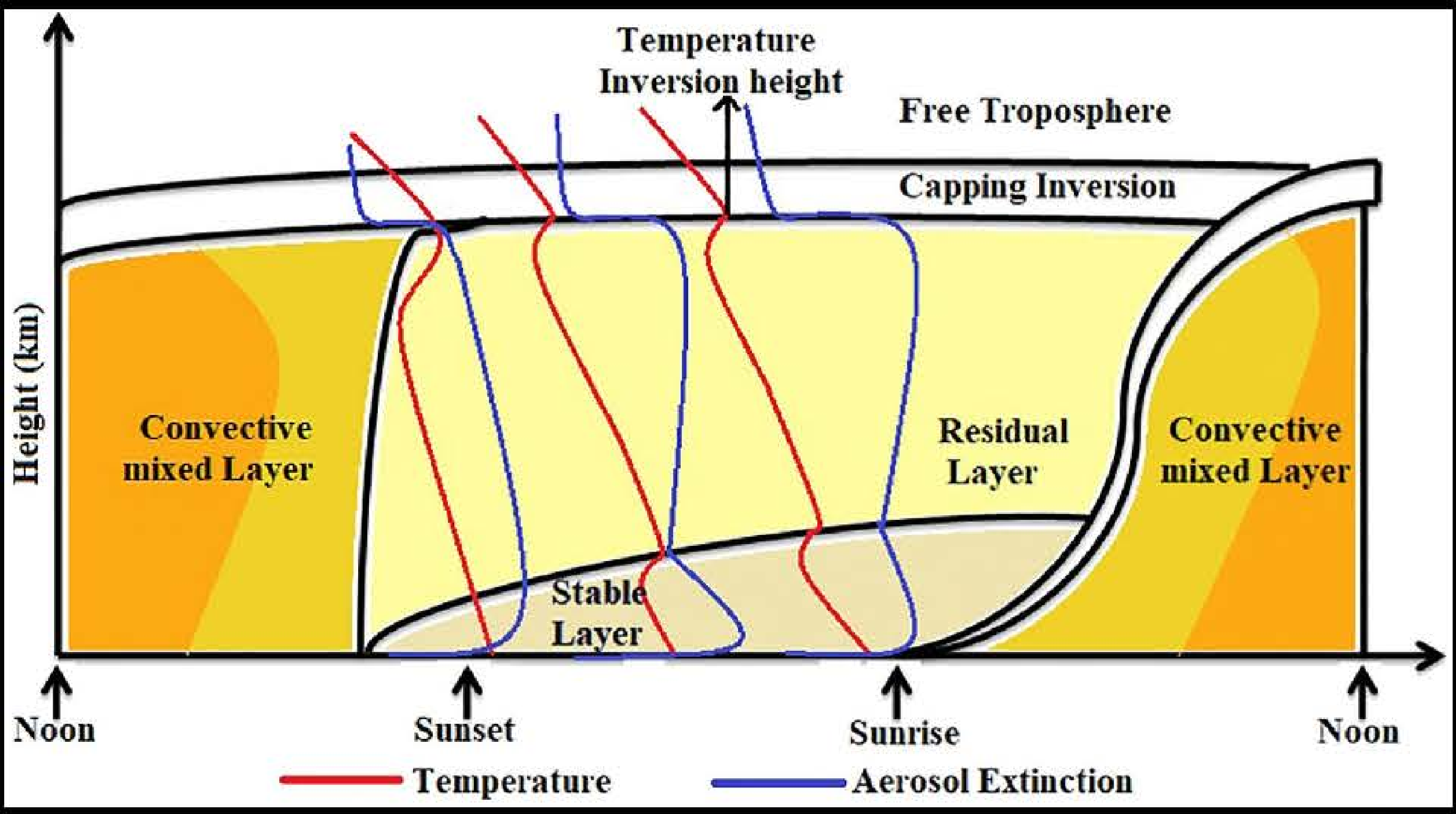

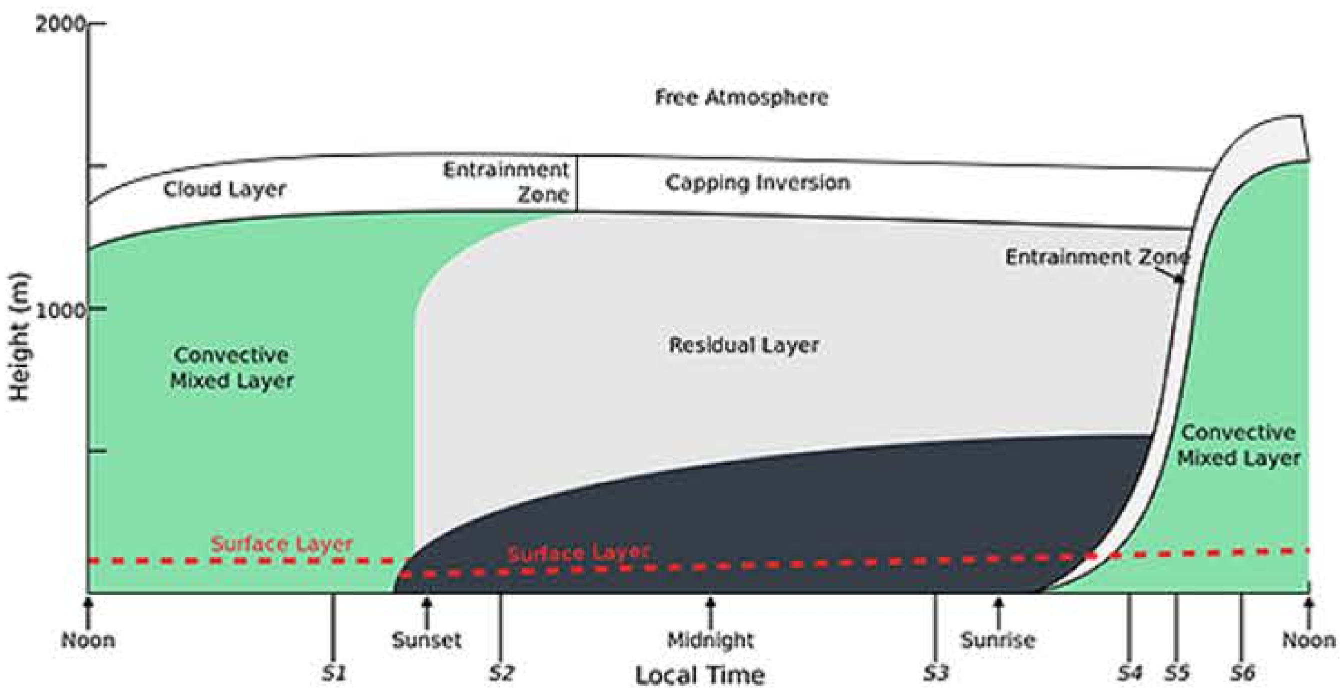

Diurnal evolution of the atmospheric boundary layer. The black region is the stable (nocturnal) boundary layer. Time markers S1–S6 are used in lesson 11.3. After R. B. Stull's An Introduction to Boundary Layer Meteorology (1988).

Summing Up

Let's summarize the diurnal behavior of the boundary layer with a bulleted list of technical terms:

Mixed Layer (Convective Boundary Layer):

- turbulence driven by convection (large eddies or thermals)
- heat transfer from solar heating of the ground to the atmosphere
- mixed layer grows by entrainment of air from above it
- virtual temperature nearly adiabatic in middle; superadiabatic (i.e., potential temperature decreases with height) near surface; subadiabatic (i.e., potential temperature increases with height) at top, where exchange of air between the ABL and the free troposphere occurs
- wind speeds are sub-geostrophic in mixed layer, crossing isobars because of turbulent drag

Surface Layer

- directly in contact with Earth's surface
- usually has vertical gradients in potential temperature, water vapor, and other quantities
- logarithmic wind speed profile with height, with low wind speed near ground
- typically is ~10% of the mixed layer

Residual Layer

- disconnected from boundary layer and Earth's surface
- neutrally stratified, with small but near-equal turbulence in all directions
- contains moisture and trace atmospheric constituents from the day before

Stable Boundary Layer

- statically stable with weaker turbulence that occurs sporadically
- winds aloft may increase to supergeostrophic speeds (low-level jet or nocturnal jet)
- stability tends to suppress turbulence, except for occasional shear-generated turbulence caused by the low-level jet

PennState College of Earth and Mineral Sciences

John A. Dutton e-Education Institute

ENTRAINMENT ZONE DEFINITIONS: A LARGE EDDY STUDY

Alison M. Fowler*, Ian M. Brooks University of Leeds, Leeds, UK

The **entrainment zone** (EZ) is the region at the top of the mixed layer where the free atmosphere above is entrained downward into the mixed layer, and thermals overshoot upward of the mixed layer. Entrainment is responsible for the growth of the boundary layer (BL). The height of the mixed layer is important for understanding the dilution and transportation of pollutants emitted from the surface and also the turbulent structure within the mixed layer under unstable conditions, due to the height of the mixed layer limiting the vertical growth of the eddies (Gryning et al., 1987). Entrainment also plays a key role in determining the distribution and structure of stratiform clouds, particularly marine stratocumulus which is an important factor affecting the global radiation budget, and hence climate.

Entrainment is a result of turbulence driven by the surface heat flux, and turbulence generated by wind shear. Thermals which are positively buoyant at the surface rise through the mixed layer until they reach the warmer free atmosphere and become negatively buoyant; they overshoot a small distance because of their momentum (Stull, 1988). The negatively buoyant air sinks back down into the mixed layer intact due to low turbulence in the free atmosphere, so pollutants from within the mixed layer stay within the mixed layer. In the process of overshooting into the free atmosphere, free atmosphere air is drawn into the mixed layer and is quickly mixed in due to the strong turbulence within the mixed layer – thus the mixed layer grows.